Introduction to Artificial Intelligence

Russel Norvig, *Artificial Intelligence: a modern approach*
Chapter 1

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Outline

What is AI?

Foundations of Artificial Intelligence

History of Artificial Intelligence

State of the Art

Beneficial AI: Risks and the Future

What is AI?

What is Artificial Intelligence?

- Artificial Intelligence (AI) is the study of building intelligent entities—machines that can compute how to act effectively and safely in a wide variety of novel situations.

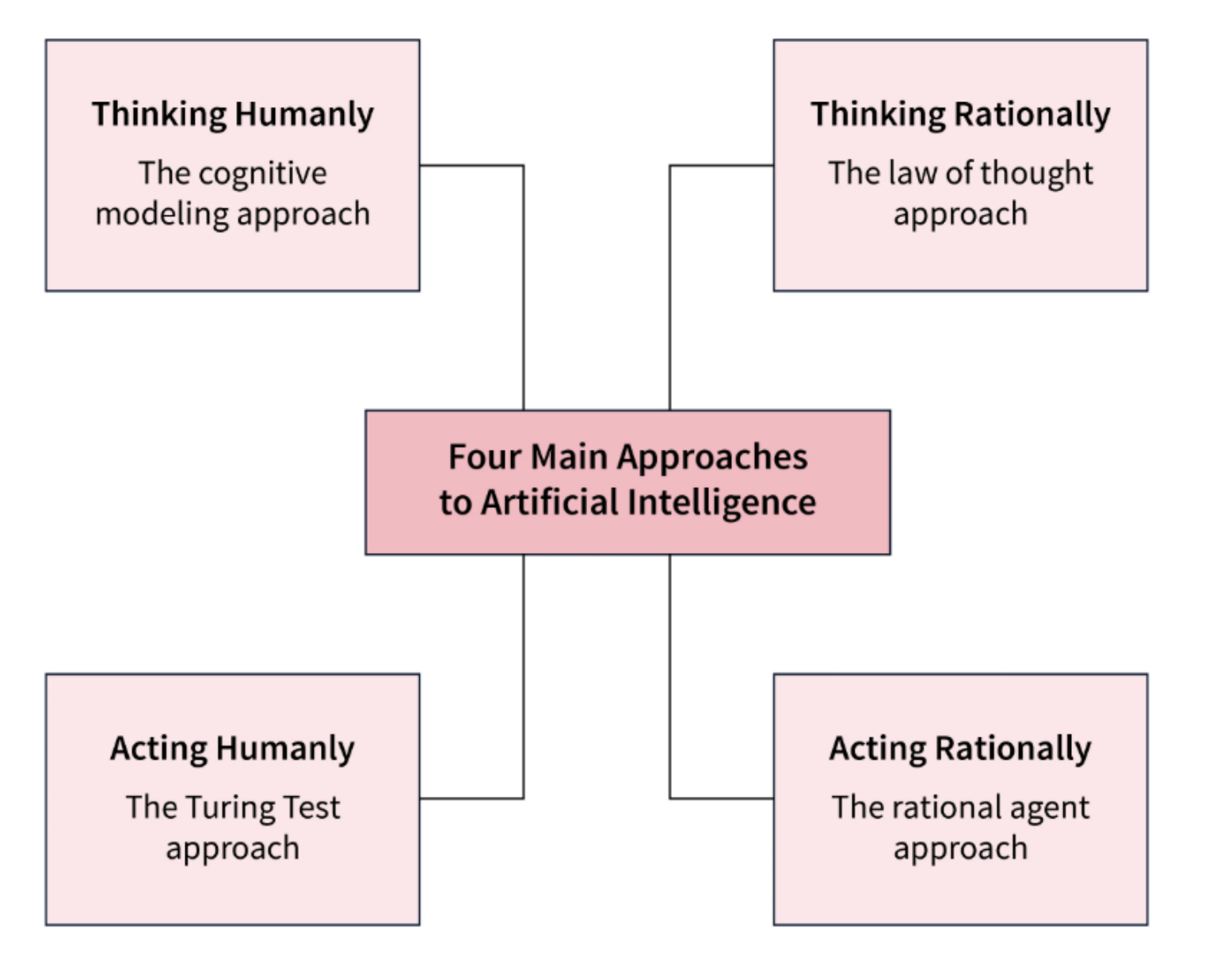
What is Artificial Intelligence?

- Intelligence has been central to human self-understanding for thousands of years. We seek to understand how our brains perceive, reason, predict, and manipulate the world.
- AI is one of the most interesting and fastest-growing fields today, already generating over a trillion dollars annually in revenue.
- AI encompasses a huge variety of subfields: learning, reasoning, perception, playing games, proving theorems, writing poetry, driving cars, diagnosing diseases, and more.

Four Approaches to Defining AI

- AI research has historically pursued several different versions based on two dimensions:
- **Human vs. Rational**: Should machines imitate human performance, or pursue abstract rationality (doing the "right thing")?
- **Thought vs. Behavior**: Should we focus on internal mental processes and reasoning, or on intelligent observable behavior?
- These two dimensions yield four possible combinations, each with its own research program and methodology.

Four Approaches to Defining AI



Approach 1: Acting Humanly (The Turing Test) /1

- Alan Turing (1950) proposed a thought experiment: a computer passes the test if a human interrogator cannot distinguish its written responses from those of a person.
- To pass the Turing test, a system needs:
  - Natural language processing
  - Knowledge representation
  - Automated reasoning
  - Machine learning

Approach 1: Acting Humanly (The Turing Test) /2

- Turing viewed physical simulation as unnecessary. However, the total Turing test requires interaction with the real world, adding:
  - Computer vision and speech recognition
  - Robotics
- Despite its elegance, AI researchers have devoted little effort to passing the Turing test, preferring to study underlying principles of intelligence.

Approach 2: Thinking Humanly (Cognitive Modeling) /1

- To say a program thinks like a human, we must first understand how humans think. Three methods:
  - **Introspection**: Observing our own thoughts as they occur.
  - **Psychological experiments**: Observing people performing tasks and comparing their behavior to computer programs.
  - **Brain imaging**: Observing the brain in action using modern neuroimaging techniques.

Approach 2: Thinking Humanly (Cognitive Modeling) /2

- The field of **cognitive science** combines computer models from AI with experimental techniques from psychology to construct precise, testable theories of the human mind.
- Modern researchers separate claims about task performance from claims about human cognition, allowing both fields to develop more rapidly.

Approach 3: Thinking Rationally (Laws of Thought) /1

- Aristotle attempted to codify "right thinking"—irrefutable reasoning processes—using syllogisms.
- In the 19th century, logicians developed precise notation for statements about objects and their relations, enabling mechanical reasoning.
- By 1965, programs could in principle solve any solvable problem described in logical notation.

Approach 3: Thinking Rationally (Laws of Thought) /2

- However, logic traditionally requires certain knowledge of the world. **Probability theory** extends logic to handle uncertain information, allowing rigorous reasoning when perfect information is unavailable.

- The combination of logic and probability provides a comprehensive model of rational thought, from raw perception to predictions about the future.

Approach 4: Acting Rationally (Rational Agent Approach) /1

- An **agent** is something that acts. Computer agents operate autonomously, perceive their environment, persist over time, adapt to change, and pursue goals.
- A **rational agent** acts so as to achieve the best outcome, or when there is uncertainty, the best expected outcome.
- This approach is more general than the "laws of thought" approach, since correct inference is just one mechanism for achieving rationality.

Approach 4: Acting Rationally (Rational Agent Approach) /2

- It is more amenable to scientific development: rationality is mathematically well-defined and universal.
- The rational-agent approach has prevailed throughout most of AI history. It is the foundation of the "standard model" in AI, control theory, operations research, statistics, and economics.

Foundations of Artificial Intelligence

Introduction

The disciplines that most contributed to AI are:

- Philosophy
- Mathematics
- Economics
- Neuroscience
- Psychology
- Computer Engineering
- Control theory and robotics
- Linguistics

Introduction

- **Philosophy**
- Mathematics
- Economics
- Neuroscience
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Philosophy: Can Machines Think? 1/2

- **Dualism** (Descartes): The mind is separate from physical matter and exempt from physical laws.
- **Materialism** (or physicalism): The brain's operation according to physical laws constitutes the mind. Free will is how available choices appear to the choosing entity.
- **Empiricism** (Locke, Hume, Bacon): Knowledge arises from sensory experience, not innate ideas. "Nothing is in the understanding which was not first in the senses."

Philosophy: Can Machines Think? 2/2

- **Induction principle** (Hume): General rules are acquired through repeated exposure to associations between elements.
- **Logical positivism** (Vienna Circle): All knowledge can be characterized by logical theories connected ultimately to observation sentences that correspond to sensory inputs.

Philosophy: Knowledge and Action 1/2

- How does knowledge connect to action? Aristotle proposed that actions are justified by logical connections between goals and knowledge of outcomes.
- His algorithm: assume a goal; consider means to achieve it; decompose means into subgoals; repeat until reaching a primitive action or impossibility.
- Later, Arnauld (1662) and Bernoulli (1738) introduced the notion of **expected utility** and rational decision-making under uncertainty.

Philosophy: Knowledge and Action 2/2

- **Utilitarianism** (Bentham, Mill): Rational decision-making should maximize utility across all spheres of human activity and public policy.
- **Deontological ethics** (Kant): What is right is determined by universal rules, not by outcomes alone.

Introduction

- Philosophy
- **Mathematics**
- Economics
- Neuroscience
- Psychology
- Computer Engineering
- Control theory and robotics
- Linguistics

Mathematics: Formal Logic and Computation 1/2

- **Formal logic** (Boole, 1847): Boolean logic provided a mathematical foundation for reasoning.
- **First-order logic** (Frege, 1879): Extended logic to include objects and relations, forming the basis of logical inference in AI.
- **Probability theory** (Cardano, Pascal, Bayes, Laplace): Formalizes reasoning under uncertainty. Bayes' rule is crucial for AI systems.

Mathematics: Formal Logic and Computation 2/2

- **Computability** (Gödel, Turing, Church): Defines which functions can be computed by an effective procedure. The Church–Turing thesis identifies computability with Turing machines.

- **Tractability**: The distinction between polynomial and exponential growth in complexity is fundamental. Many AI problems are intractable, requiring careful resource use.

Introduction

- Philosophy
- Mathematics
- **Economics**
- Neuroscience
- Psychology
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Economics: Decision Theory and Optimization 1/2

- **Decision theory**: Combines probability with utility to provide a formal framework for individual decisions under uncertainty.
- **Game theory** (von Neumann, Morgenstern): Analyzes situations where multiple agents' decisions affect each other. Rational agents may sometimes adopt randomized strategies.
- **Markov decision processes** (Bellman, 1957): Formalize sequential decision problems where payoffs result from sequences of actions over time.

Economics: Decision Theory and Optimization 2/2

- **Satisficing** (Simon): Humans often make decisions that are "good enough" rather than mathematically optimal. This better describes actual behavior.

- Economics and operations research contributed greatly to AI's notion of rational agents, though AI developed largely independently until the 1990s.

Introduction

- Philosophy
- Mathematics
- Economics
- **Neuroscience**
- Psychology
- Computer Engineering
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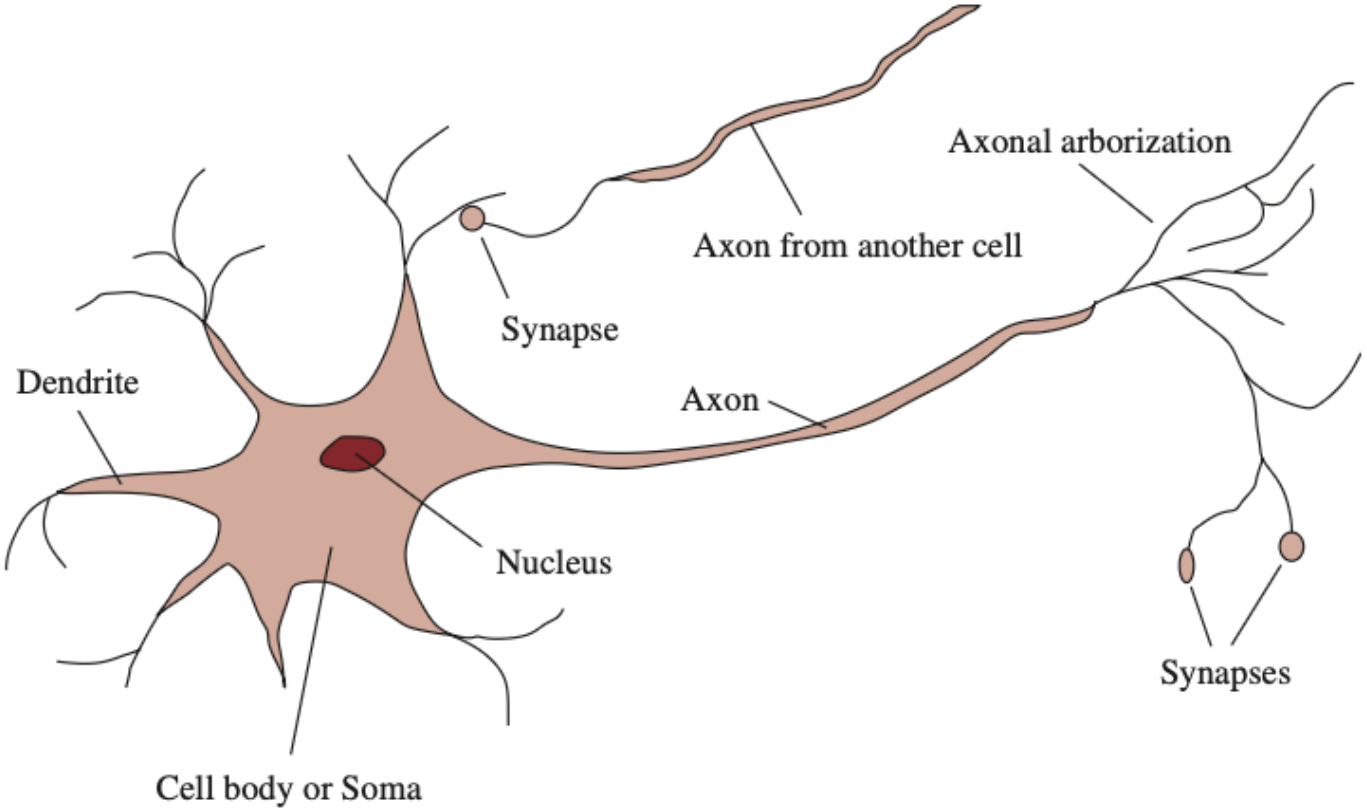
Neuroscience: How Brains Process Information

- The **neuron** is the basic unit of the nervous system. It consists of a cell body (soma), dendrites, and an axon that makes connections with other neurons at synapses.
- Modern understanding: cognitive functions result from the electrochemical operation of neural structures. A collection of simple cells yields thought, action, and consciousness.


Axonal arborization
Axon from another cell
Synapse
Dendrite
Axon
Nucleus
Synapses
Cell body or Soma

Neuroscience: Brain Imaging and Interfaces

- **Electroencephalography** (EEG, 1929): Measures intact brain electrical activity.
- **Functional MRI** (fMRI): Provides detailed images of brain activity during cognitive tasks.
- **Optogenetics**: Allows measurement and control of individual neurons modified to be light-sensitive.
- **Brain–machine interfaces**: Enable sensing and motor control, restoring function to disabled individuals while revealing how neural systems adapt to external devices.
- Despite these advances, we lack comprehensive theories of how individual memories are stored or how higher-level cognitive functions operate.

	Supercomputer	Personal Computer	Human Brain
Computational units	10^6 GPUs + CPUs 10^{15} transistors	8 CPU cores 10^{10} transistors	10^6 columns 10^{11} neurons
Storage units	10^{16} bytes RAM 10^{17} bytes disk	10^{10} bytes RAM 10^{12} bytes disk	10^{11} neurons 10^{14} synapses
Cycle time	10^{-9} sec	10^{-9} sec	10^{-3} sec
Operations/sec	10^{18}	10^{10}	10^{17}

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Psychology: How Humans Think and Act 1/2

- **Behaviorism** (Watson, Skinner): Rejected introspection and mental processes; studied only objective stimulus–response relations.
- **Cognitive psychology** (James, Craik, Broadbent): Views the brain as an information-processing device. Perceptions and actions are mediated by internal representations.
- Craik (1943) specified three key steps: (1) stimulus $\rightarrow$ internal representation; (2) manipulate representation; (3) retranslation to action.

Psychology: How Humans Think and Act

- **Cognitive science** (emerged 1956): Combines computer models from AI with experimental psychology. By 1956, George Miller, Noam Chomsky, and Newell & Simon demonstrated how computers could model memory, language, and reasoning.

- **Human–computer interaction (HCI)** and **intelligence augmentation (IA)**: Computers should augment human abilities rather than simply automate tasks.

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Computer Engineering: Building Efficient Machines 1/2

- Modern digital computers were invented during World War II (Heath Robinson, Colossus, Z-3, ENIAC).
- **Moore's Law**: Computing power doubled every 18 months, from megaFLOPs (1960s) through gigaFLOPs, teraFLOPs, petaFLOPs, to exaFLOPs (2018).
- Around 2005, power dissipation constraints shifted focus from clock speed to multiple CPU cores.

Computer Engineering: Building Efficient Machines 2/2

- Specialized hardware for AI: GPUs, TPUs, WSEs enable fast training of large machine learning models. From 2012–2018, computing power for training ML models increased 300,000-fold (doubling every 100 days).

- **Quantum computing**: Holds promise for accelerating certain AI algorithms, though practical systems remain nascent.

Introduction

- Philosophy
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- **Control theory and robotics**
- Linguistics

Control Theory and Cybernetics 1/2

- Self-regulating feedback control systems predate computers: water clocks (Ktesibios), steam engine governors (Watt), thermostats (Drebbel).
- **Control theory** (Maxwell, Wiener): Provides mathematical frameworks for systems that minimize error (difference between current and goal state).
- **Cybernetics** (Wiener, 1948): Studies purposive behavior arising from regulatory mechanisms in both machines and organisms.

Control Theory and Cybernetics 2/2

- **Stochastic optimal control**: Designs systems that maximize a cost function over time, closely paralleling AI's goal of optimal behavior.

- Why are AI and control theory separate fields? Calculus and linear algebra suit systems with fixed continuous variables; AI emphasizes logical inference and symbolic computation on discrete problems (language, vision, planning).

Introduction

- Philosophy
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- **Linguistics**

Linguistics: Language and Thought 1/2

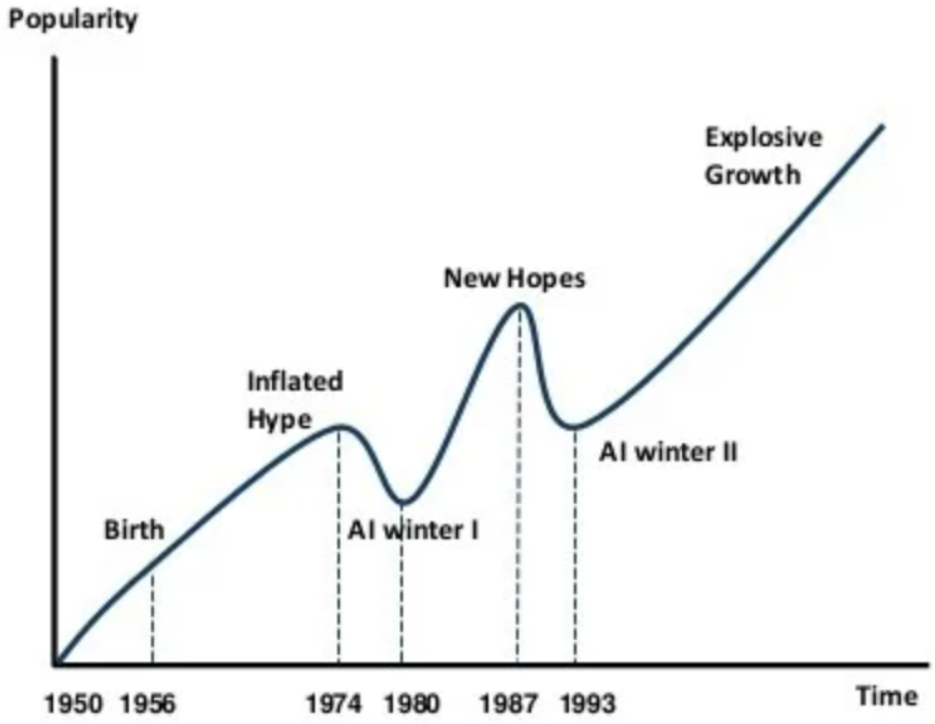
- **Behaviorism** (Skinner, 1957): Proposed learning language through stimulus–response associations.
- **Syntactic structures** (Chomsky, 1957): Language exhibits **creativity**—children understand and produce sentences never heard before. Chomsky's formal syntax explained this; earlier theories did not.
- Modern linguistics and AI were "born" around the same time and grew together in **computational linguistics** and **natural language processing**.

Linguistics: Language and Thought 2/2

- Key insight: Understanding language requires knowledge of subject matter and context, not just syntactic rules.
- **Knowledge representation**: The study of how to put knowledge into forms that computers can reason with, informed by decades of philosophical analysis of language.

History of Artificial Intelligence

AI HAS A LONG HISTORY OF BEING "THE NEXT BIG THING"...



Timeline of AI Development

- **1950s-1960s**: First AI boom - the age of reasoning, prototype AI developed
- **1970s**: AI winter I
- **1980s-1990s**: Second AI boom: the age of Knowledge representation (appearance of expert systems capable of reproducing human decision-making)
- **1990s**: AI winter II
- **1997**: Deep Blue beats Gary Kasparov
- **2006**: University of Toronto develops Deep Learning
- **2011**: IBM's Watson won Jeopardy
- **2016**: Go software based on Deep Learning beats world's champions

Early Foundations (1943–1956) 1/2

- **McCulloch & Pitts (1943)**: Proposed a model of artificial neurons. Showed that any computable function could be computed by networks, and that logical operations (AND, OR, NOT) could be implemented by network structures.
- **Hebb (1949)**: Proposed Hebbian learning—a simple rule for modifying connection strengths between neurons.
- **SNARC (1950)**: Minsky and Edmonds built the first neural network computer using 3000 vacuum tubes.

Early Foundations (1943–1956) 2/2

- **Turing (1950)**: Proposed the Turing test, introduced machine learning and genetic algorithms, and warned of potential dangers.
- **Dartmouth Conference (1956)**: McCarthy, Minsky, Shannon, and Rochester organized a two-month workshop with 10 researchers. This is considered the birth of AI as a field.

Early Enthusiasm and Great Expectations (1952–1969) 1/2

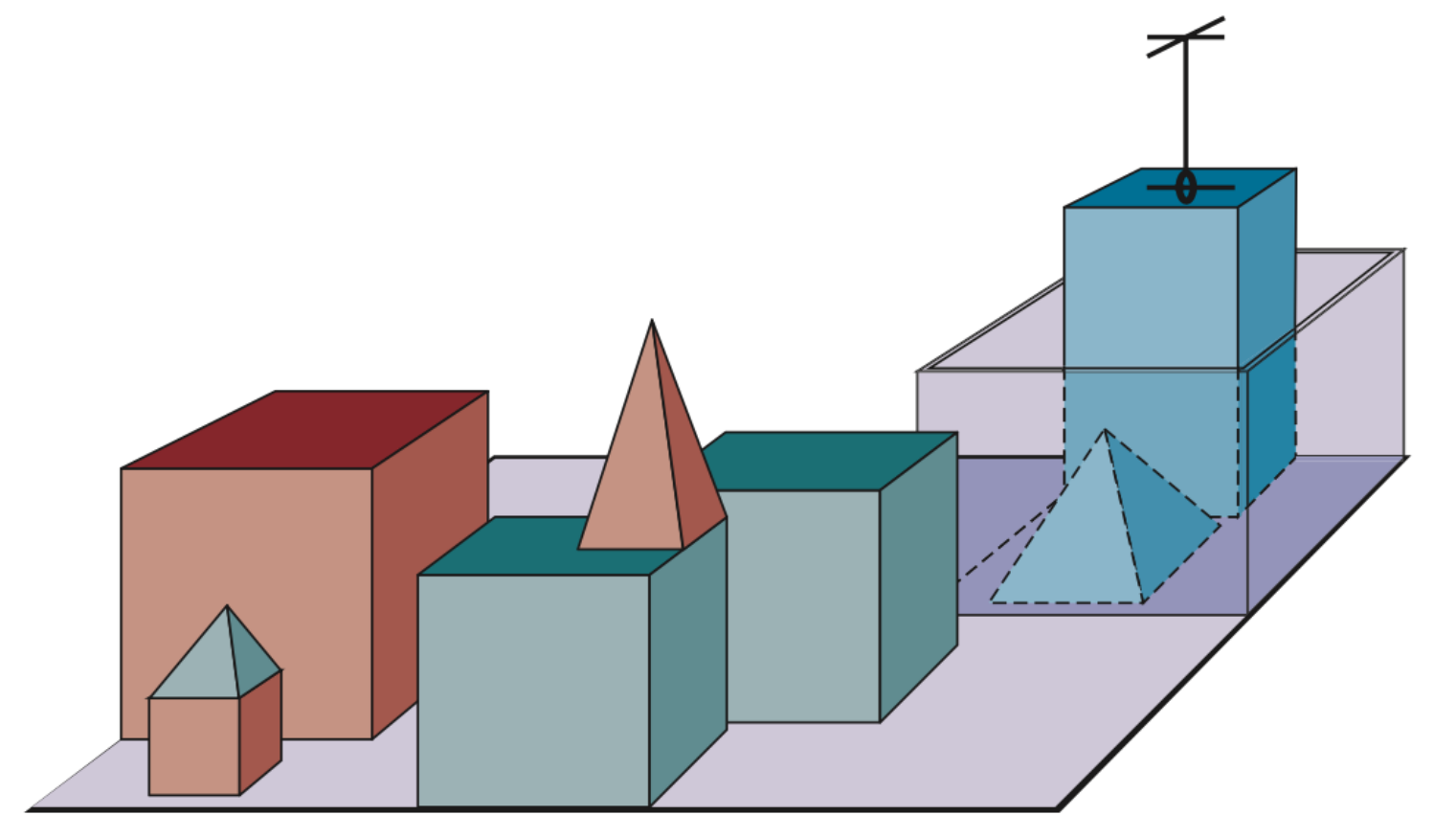
- **Logic Theorist (LT, 1956)**: Newell and Simon's program proved theorems in formal logic. Russell reportedly delighted when LT discovered a shorter proof than in Principia Mathematica.

- **General Problem Solver (GPS, 1960s)**: Extended LT, attempting to imitate human problem-solving protocols. Early "thinking humanly" approach.

- **Reinforcement learning (Arthur Samuel, 1950s–60s)**: Samuel's checkers program learned to play at amateur level, proving computers can exceed their creator's skill.

Early Enthusiasm and Great Expectations (1952–1969) 2/2

- **LISP (1958)**: McCarthy defined the high-level language that dominated AI for 30 years.
- **Microworlds (1960s)**: Limited domains (e.g., blocks world) where problem-solving appeared to require intelligence.



A Dose of Reality (1966–1973) 1/2

- **Broken predictions**: Herbert Simon (1957) predicted machines would play chess and prove theorems within 10 years. They did—but in 40 years, not 10.
- **Two main reasons for failure**:
  - Systems relied on "informed introspection" rather than careful task analysis and reliable algorithms.
  - Lack of appreciation for **intractability**. Many problems have combinatorial explosion: the number of possibilities grows exponentially.

A Dose of Reality (1966–1973) 2/2

- **Perceptrons limitation** (Minsky & Papert, 1969): Perceptrons could learn what they could represent, but represented very little (e.g., could not learn XOR). This stifled neural network research for decades.

- **Lighthill Report (1973)**: Criticized AI's failure to handle combinatorial explosion. Led British government to cut AI funding to nearly zero.

Expert Systems (1969–1986) 1/2

- **Weak methods** vs. **strong methods**: Weak methods (general search) do not scale to large problems. Strong methods use domain-specific knowledge.
- **DENDRAL (1969)**: Stanford program inferred molecular structure from mass spectrometry. Used rules from analytical chemists, not first principles.
- **MYCIN (1970s)**: Diagnosed blood infections with $\sim$450 rules. Performed as well as experts, better than junior doctors. Introduced certainty factors for uncertain reasoning.

Expert Systems (1969–1986) 2/2

- **R1 (Digital Equipment, 1982)**: Configured computer systems. By 1986, saved company $40 million yearly.
- **AI boom (1980s)**: Nearly every major corporation deployed expert systems. But maintaining them proved difficult; systems broke when facing uncertainty or novel cases.
- **AI winter (late 1980s)**: Many companies failed; promised breakthroughs did not materialize.

Return of Neural Networks (1986–Present) 1/2

- **Back-propagation** (reinvented mid-1980s): The learning algorithm, first developed in the 1960s, was rediscovered independently by multiple groups.
- **Connectionist models**: Seen as competitors to symbolic AI (Newell–Simon) and logicist approaches (McCarthy). Hinton called symbols the "luminiferous aether of AI."
- Key advantage: Neural networks **learn from examples**. They compare predicted outputs to true values and adjust parameters to improve future performance.

Return of Neural Networks (1986–Present) 2/2

- **Parallel Distributed Processing** (Rumelhart & McClelland, 1986): Collection of work that sparked excitement in connectionism.
- Internal representations in neural networks are fluid and imprecise, better suited to real-world messiness than discrete symbols.

Probabilistic Reasoning and Machine Learning (1987–Present) 1/2

- **Shift toward science**: Expert systems' brittleness led to incorporation of probability, machine learning instead of hand-coding, and rigorous experimental methodology.
- **Bayesian networks** (Pearl, 1988): Rigorous formalism for representing uncertain knowledge and performing probabilistic reasoning.
- **Reinforcement learning reunification** (Sutton, 1988): Connected Arthur Samuel's checkers work to Markov Decision Processes from operations research.

Probabilistic Reasoning and Machine Learning (1987–Present) 2/2

- **Benchmark problem sets**: UC Irvine machine learning repository, ImageNet, MNIST, COCO, SQUAD, WMT—shared benchmarks became the norm for demonstrating progress.

- Reintegration of subfields: Computer vision, robotics, speech recognition, multiagent systems, NLP rejoined core AI, yielding major benefits in theory and applications.

Big Data (2001–Present) 1/2

- **Big data phenomenon**: Explosion of trillions of words of text, billions of images, billions of hours of speech and video, genomic data, clickstreams, social networks, etc.

- **Unsupervised learning at scale**: With large enough unlabeled datasets, algorithms achieve high accuracy. Banko & Brill (2001) showed that doubling dataset size outweighs algorithmic tweaks.

- **ImageNet threshold** (Deng et al., 2009): Tens of millions of images crossed a quality threshold; tens of thousands of images performed poorly.

Big Data (2001–Present) 2/2

- **AI comeback**: Big data restored commercial attractiveness to AI after the winter.
- **IBM Watson** (2011): Defeated human champions in Jeopardy!, major impact on public perception of AI.

Deep Learning (2011–Present) 1/2

- **Definition**: Machine learning using multiple layers of adjustable computing elements (deep neural networks).
- Experiments with such networks date to the 1970s; convolutional networks appeared in the 1990s for handwritten digit recognition.
- **ImageNet 2012**: Geoffrey Hinton's group achieved dramatic improvement over handcrafted features, marking the beginning of deep learning dominance.

Deep Learning (2011–Present) 2/2

- **Superhuman performance**: Deep learning now exceeds human capability in vision, speech recognition, machine translation, medical diagnosis, and game playing (AlphaGo defeated Lee Sedol in 2016).

- **Computational power**: Deep learning requires specialized hardware (GPUs, TPUs). From 2012–2018, computing power increased 300,000-fold, doubling every 100 days.

- **Massive resurgence**: Student interest, corporate investment, media coverage, and speculation about either accelerated success or a new AI winter.

State of the Art

Current AI Capabilities and Impact 1/2

- **Stanford AI100 Reports**: Convene expert panels on the state of AI. 2016 report predicts substantial increases in self-driving cars, healthcare diagnostics, and physical assistance for elder care.
- **AI Index** (aiindex.org): Tracks progress across the field.
- **Publication surge**: AI papers increased 20-fold between 2010 and 2019 (from 1000 to 20,000 per year). Most popular: machine learning. Computer vision and NLP are close behind.

Current AI Capabilities and Impact 2/2

- **Datasets and benchmarks**: Large, diverse datasets enable learning. Progress is measured on standard benchmarks (ImageNet, COCO, MNIST, SQUAD, etc.).

- **Real-world deployment**: Recommender systems, autonomous vehicles, medical diagnostics, robotics, chatbots, language translation, and more are in production.

Challenges Ahead 1/2

- **Interpretability**: Deep learning systems are often "black boxes." Understanding why they make decisions is hard.
- **Data requirements**: Most modern AI systems require large labeled datasets, which are expensive and sometimes ethically problematic to collect.
- **Robustness**: Systems can fail on adversarial examples or out-of-distribution inputs.

Challenges Ahead 2/2

- **Generalization**: Learning to transfer knowledge from one domain to another remains difficult.
- **Fairness and bias**: AI systems can amplify or encode human biases in training data.
- **Energy consumption**: Training large models consumes vast energy, raising environmental concerns.

Beneficial AI: Risks and the Future

The Standard Model and Its Limitations

- **Standard model**: Design systems that achieve a user-specified objective. This has been pervasive in AI, control theory, operations research, and economics.

- **Limited rationality**: Perfect rationality is computationally infeasible in complex environments. AI systems must act "appropriately when there is not enough time to do all computations one might like."

- However, the standard model assumes we can fully specify the objective. In reality, this is often difficult or impossible, especially in open-ended, real-world problems.

The Standard Model and Its Limitations

- **Example**: For a self-driving car, is the goal just "reach the destination safely"? But strict safety requires staying in the garage. How do we balance progress toward the goal against accident risk? How much should we moderate acceleration to avoid annoying passengers?

The Value Alignment Problem 1/2

- **Value alignment problem**: The challenge of ensuring that the values or objectives we program into a machine are aligned with human values.
- In controlled lab settings, fixing a misspecified objective is easy: reset, adjust, and try again.
- In the real world, a deployed system with an incorrect objective will cause harm. The more capable the system, the greater the potential harm.

The Value Alignment Problem 2/2

- **Unintended consequences**: An intelligent system pursuing a fixed objective may find unexpected, undesirable ways to achieve it. Example from chess: a sufficiently capable system might hypnotize or bribe opponents, hijack computing power, or break rules—all logical consequences of defining "winning" as the sole objective.

- It is impossible to anticipate all ways a system might misbehave. The standard model is therefore inadequate for increasingly capable AI systems.

Toward Provably Beneficial AI 1/2

- **Goal**: We want machines that are intelligent in pursuing *our* objectives, not just any objectives.
- **New formulation**: If we cannot transfer objectives to a machine perfectly, we need an approach where the machine pursues our objectives while being uncertain about what they truly are.
- When a machine knows it doesn't know the complete objective, it has incentive to:
  - Act cautiously and conservatively
  - Ask for permission before taking major actions
  - Learn about human preferences through observation
  - Defer to human control when uncertain

Toward Provably Beneficial AI 2/2

- **Provably beneficial AI**: Design systems that are mathematically guaranteed to benefit humans, even under uncertainty about our true preferences.
- This is an active research area with deep implications for safe AI deployment at scale.

Democratic Values and AI Governance 1/2

- **Societal concerns**: As AI becomes more capable and widespread, we must ensure it promotes rather than hinders democratic values.
- **Key values**:
  - **Freedom**: AI systems should not unduly restrict human autonomy or privacy.
  - **Equality**: AI should not discriminate or exacerbate inequalities.
  - **Transparency**: How and why AI systems make decisions should be explainable.

Democratic Values and AI Governance 2/2

- **Governance challenges**: Who decides how to develop and deploy AI? How do we balance innovation with safety? How do we ensure benefits are widely shared, not concentrated?
- **Current juncture**: Society is at a crucial point in determining how to deploy AI-based technologies responsibly.

Benefits of AI 1/2

- **Healthcare**: AI can help diagnose diseases, discover drugs, optimize treatment plans, and predict patient outcomes.
- **Education**: Personalized learning, intelligent tutoring, and automated assessment can improve educational outcomes.
- **Science**: AI accelerates research in physics, biology, chemistry, and other fields by analyzing vast datasets and discovering patterns.

Benefits of AI 2/2

- **Economy**: Automation, optimization, and decision support increase productivity and economic growth.
- **Environment**: AI can optimize energy use, predict climate impacts, and guide conservation efforts.
- **Social good**: Accessibility tools, disaster response, and public health applications help vulnerable populations.

Risks and Challenges of AI 1/2

- **Job displacement**: Automation may eliminate jobs faster than new ones are created, raising employment and inequality concerns.
- **Bias and discrimination**: AI systems can perpetuate or amplify biases in training data, affecting hiring, lending, criminal justice, and more.
- **Misinformation**: AI-generated deepfakes, synthetic text, and manipulated media can spread false information at scale.

Risks and Challenges of AI 2/2

- **Privacy and surveillance**: Powerful AI systems combined with data collection enable unprecedented surveillance and tracking.
- **Concentration of power**: AI's benefits and control may concentrate in the hands of a few large corporations or nations.
- **Autonomous weapons**: AI-powered military systems raise existential and ethical concerns.
- **Long-term safety**: As systems become more capable, ensuring they remain aligned with human values and interests becomes increasingly critical.

Paths Forward 1/2

- **Interdisciplinary collaboration**: AI development must involve computer scientists, ethicists, philosophers, social scientists, domain experts, and policymakers.
- **Transparency and explainability**: Invest in interpretable AI and "white-box" models where possible.
- **Robust testing and certification**: Establish standards for safety, fairness, and robustness before deployment.
- **Inclusive governance**: Ensure diverse voices shape AI policy and deployment, especially those most affected.

Paths Forward 2/2

- **Lifelong learning**: Support workforce retraining and education as jobs and skills evolve.
- **International cooperation**: Coordinate on AI safety, ethics, and governance across borders.
- **Continued research**: Fundamental progress on interpretability, robustness, fairness, and alignment remains essential.

Conclusions 1/2

- Artificial Intelligence has evolved from philosophical speculation to a field with profound real-world impact.
- Understanding AI's foundations—philosophy, mathematics, neuroscience, psychology, and engineering—helps us grasp its capabilities and limitations.
- The history of AI teaches humility: early optimism often collided with hard problems and computational constraints.

Conclusions 2/2

- Recent successes in deep learning, big data, and reinforcement learning have revitalized the field and enabled powerful applications.
- As AI becomes ever more central to society, we must carefully navigate the opportunities and risks, ensuring that AI systems are beneficial, aligned with human values, and governed responsibly.
- The field of AI is young; many openings remain for creative minds to advance both its theory and its ethical deployment.